

3D-to-2D Animation Smear Effect Technique Based on Japanese Hand-Drawn Animation Style

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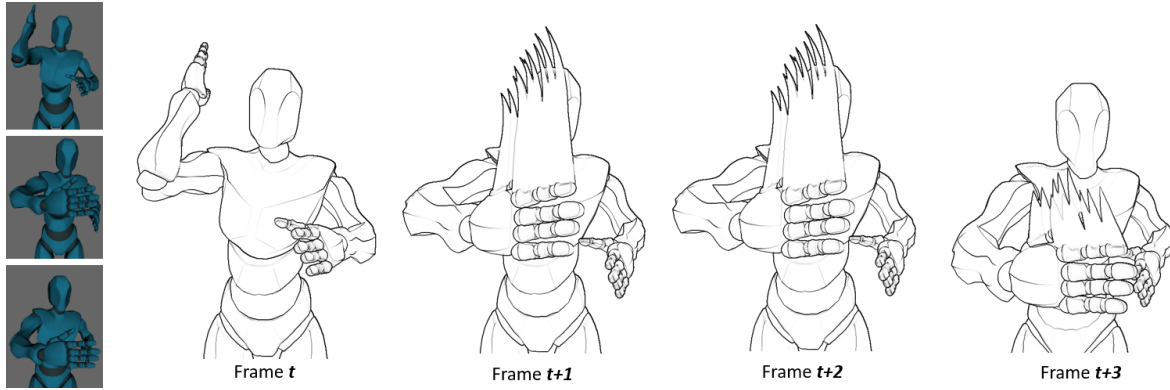


Figure 1: The proposed method is applied to a 3D model to achieve the smear effect found in Japanese hand-drawn animation. The three images on the leftmost side show the key 3D poses before applying the effect. Frames t to $t+3$ show the result after applying the effect and rendering the edges. Due to our method's consideration of the "Animating on Ones, Twos, or Threes" technique used in Japanese hand-drawn animation, Frames $t+1$ and $t+2$ display the same result.

Abstract

Smear effects are a visualization of 2D motion blur. In Japanese animation, because each keyframe of fast-moving objects might remain on the screen for more than one frame, significant differences often appear between consecutive frames, resulting in a choppy effect. To eliminate this, smear effects play an important role. In Japanese hand-drawn animation, this technique creates jagged outlines at the edges of fast-moving objects. This not only eliminates the choppy effect but also emphasizes the intensity and speed of the object's movement. The proposed method is specifically designed to accommodate the common hand-drawn technique of "Animating on Ones, Twos, and Threes" [Ryan 2021], which often appears alongside the smear effect. Our method is based on skeletal animation techniques, combined with a hierarchical processing mechanism to displace model vertices. Unlike other methods that only displace vertices in a specific direction, our method includes a bending mechanism to represent curved motion trajectories. Additionally, our method allows users to freely choose which vertices in the model to apply the effects to. Finally, the deformed model is rendered from 3D to 2D, achieving this Japanese animation-style smear effect.

Keywords

3D-to-2D, Geometric Deformation, Skeletal Animation

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1 Introduction

Toon Shading, or Cel Shading [Barla et al. 2006], is a 3D-to-2D rendering technique commonly used in animation production [Bauer and Salvati 2022] and game graphics. However, the smear technique, frequently employed in Japanese hand-drawn 2D animation, has not been well integrated into this technology. As a result, when an object is in a state of intense motion, merely applying Toon Shading and Motion Blur without artists making appropriate vertex displacements for each keyframe [Motomura 2015] can lead to a final rendering that feels discordant. Additionally, current Toon Shading methods do not take into account the concept of "Animating on Ones, Twos, and Threes" [Ryan 2021] in animation production. "Ones" means a keyframe stays for one frame, "twos" for two frames, and so on. Due to the characteristics of Japanese animation, objects in intense motion often use animation on 2's or 3's to reduce the number of drawings and accelerate production time. However, using fewer frames to depict fast-moving objects can lead to a severe choppy effect. In this context, the smear effect plays a crucial role. The smear effect, similar to hand-drawn motion blur, not only mitigates this choppy effect but also emphasizes the

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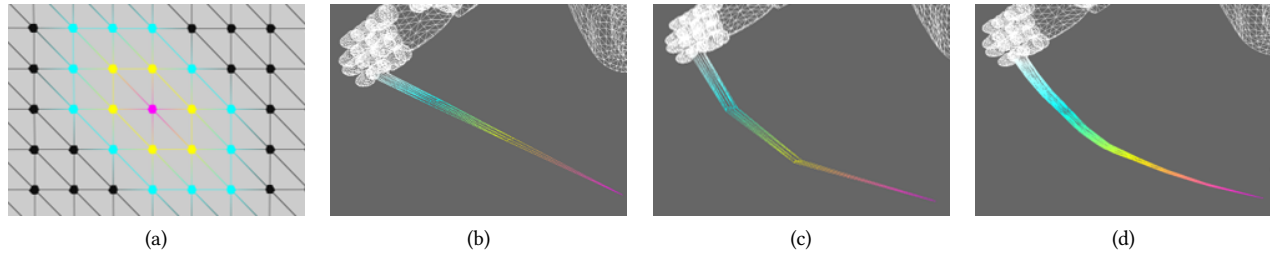


Figure 2: The results of each step in the proposed method. (a) A vertex group with three hierarchical levels. (b) The outcome after vertex group displacement. (c) Bending and curve fitting. (d) Applying smooth subdivision to (c).

sense of speed and force. This production method has become a distinctive style of Japanese hand-drawn animation. Compared to these animation production techniques, using recently emerging neural network frame interpolation technology [Bao et al. 2019], using particles [Haller et al. 2004] [Strothotte and Schlechtweg 2002], or merely displacing vertices in a certain direction [Kawagishi et al. 2003], do not effectively achieve the smear effect that depicts fast-moving objects in Japanese hand-drawn animation.

2 DESIGN AND IMPLEMENTATION

Our method primarily consists of four steps: (1) Level Grouping, (2) Vertex Displacement, (3) Bending, and (4) Smoothing. After all the steps, the result will have their length adjusted using noise (See Figure 3), and the relevant control parameters will be exposed for users to adjust. Since this method uses skeletal animation as the basis for the necessary calculations, it can be seamlessly integrated into current mainstream animation production workflows.

- (1) **Level Group** - This step provides a guidance for deformation. Users can manually select the vertices to which they want to apply the effect and set a number of hierarchical levels. Subsequently, each vertex will be transformed into vertex groups according to the number of hierarchical levels. As shown in Figure 2(a), in a vertex group with three hierarchical levels, the red vertex represent level 1, the yellow vertices represent level 2, and so on.
- (2) **Vertex Displacement** - After grouping, the skeletal movement information from the skeletal animation is used to displace each vertex group towards their positions in the previous frame. Figure 2(b) shows the displacement result for a vertex hierarchy level of 3.
- (3) **Bending** - Due to the possibility of curved motion between two frames with significantly different poses, further interpolation of the skeletal poses of the previous and next frames is performed if needed to obtain more information about the curved motion trajectory. Based on this information, a spline [Catmull and Rom 1974] is created. Then, the vertices in each hierarchical level of the vertex groups are fitted to this spline, as shown in Figure 2(c).
- (4) **Smoothing** - Due to the insufficient number of vertices, the result after bending may have noticeable sharp edges, as shown in Figure 2(c). These sharp edges will be subdivided and smoothed [Boubekeur and Alexa 2008] in this step, resulting in a smooth and curved outcome, as shown in Figure 2(d).



Figure 3: The comparison of streak lengths before applying noise (left) and after applying noise (right).

3 Conclusion AND Future Works

Our proposed smear effect method not only achieves vertex displacement effects that other methods cannot, but also allows for smooth vertex bending, providing greater flexibility to users. This makes the 3D-to-2D rendering results more closely resemble the smear effects found in Japanese hand-drawn animation. In the future, we plan to reduce the user's operational workload by developing an automated model deformation method. Furthermore, we will improve the current displacement method to reduce its coupling with the model's topological resolution.

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